

Bella Kim

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EDUCATION

University of Waterloo

Sep. 2025 – Apr. 2030

Bachelor of Applied Science in Computer Engineering

Waterloo, ON

- **Relevant Coursework:** Data Structures, Algorithms, Probability & Statistics, Linear Algebra, Computer Systems
- **Awards:** President's Scholarship of Distinction

EXPERIENCE

Graze AI

Jan. 2026 – Apr. 2026

Software Engineering Co-op

Remote

- **Architected an asynchronous FastAPI backend** integrating LLM and web-scraping APIs, utilizing `asyncio.Semaphore` to orchestrate 30+ concurrent HTTP requests while strictly managing token rate limits.
- **Engineered a deterministic entity resolution algorithm** implementing fuzzy stem matching, Levenshtein edit distances, and custom heuristic scoring to accurately map unstructured search data to canonical domains.
- **Developed an automated classification pipeline** using LLMs to parse natural language criteria into strict Pydantic schemas, with fallback deterministic logic to ensure zero-hallucination data mapping.
- **Implemented a high-throughput caching layer** utilizing `aiosqlite` and SHA-256 fingerprinting, optimizing state management and deduplicating pipeline operations across large-scale CSV datasets.

PROJECTS

Autonomous Firefighter Bot | C++, TraxMaker, AutoCAD, Embedded Systems

Feb. 2025 – Jun. 2025

- **Engineered an autonomous real-time system** in C++ to process sensor telemetry, detect anomalies, and execute deterministic pathfinding algorithms within a complex physical maze environment.
- Achieved the **fastest algorithm execution and physical completion time (9.8s)** among 50 competing robots.
- **Designed and assembled 3 custom PCBs** using TraxMaker to minimize electrical interference, improving hardware-to-software signal reliability and overall system latency by 40%.

UW Eng Network | Python, FastAPI, SQLite, ETL, Asynchronous I/O

Mar. 2026

- **Built a low-latency, full-stack architecture** integrating strict JWT authentication protocols and automated visual verification logic utilizing Claude's vision API endpoints.
- **Engineered an async web scraping ETL pipeline** (`httpx`, `BeautifulSoup`) to extract, transform, and load 300+ nodes of unstructured faculty data into a scalable SQLite database.

Hase Data Pipeline | TypeScript, React Native, SQL

Apr. 2026 – Present

- **Architecting a highly concurrent financial tracking client** capable of aggregating and logging daily income and expense data streams with real-time dynamic visualization.
- **Engineered a centralized relational database schema** to securely parse, store, and retrieve digital wallet identifiers and loyalty point metrics across mobile ecosystems.

SKILLS

Core Languages : C++, Python, SQL, TypeScript, JavaScript, Verilog

Systems & Hardware : Embedded Systems, FreeRTOS, Arduino, PCB Design, AutoCAD, TraxMaker

Frameworks & Tools : FastAPI, Flask, React, Next.js, Git, SQLite, Asynchronous I/O, ETL Pipelines